MAT3373: Long Homework 1

September 6, 2026

Homework Policy Basics

1. You can find the homework deadline in the course schedule on Brightspace.
2. Homework should be submitted in two parts: an easily read file in PDF or HTML format, and runnable code, usually in an `.ipynb` file.
3. Note: we are using Python this term. other languages such as R are also allowed, but some function names and implementation details may differ.
4. Use a fixed random seed when a question involves simulation or a random split. Before submitting a notebook, restart the kernel and run all cells from top to bottom. Marks may be taken off if the submitted code does not run.
5. Whenever preprocessing and cross-validation are both used, fit the preprocessing separately within each training fold. Whenever a final test set is used, fit preprocessing using only the development or training data.
6. Homework solutions should appear in the same order as the questions in this document.
7. It is possible to submit several versions of the homework. I will grade the last version submitted before the deadline. I strongly encourage you to submit drafts as you go.

Unless a question explicitly says otherwise, "logistic regression" means unpenalized logistic regression. Named functions and `random_state` values refer to Python. If you use R, use the corresponding method and report the random seed and any software defaults that affect your answer.

The instructions are fairly long. This does not mean your answers should be long. A short answer that includes the relevant calculation, plot, or explanation is usually better than a long answer that does not directly address the question.

I expect to grade only a subset of the parts. Several supplied datasets are synthetic and deliberately exaggerate the relevant phenomenon; they should not be treated as realistic scientific models.

The Python package `statsmodels` is useful in Questions 4 and 5, where you are asked for ordinary regression confidence intervals or p-values.

1 Training Error and New-Data Error for a Constant Predictor

Let $Y_1, \ldots, Y_n$ be independent observations with mean μ and variance σ^2. Ignore any predictors and fit the constant prediction rule

$$\widehat{f}(x) = \overline{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i.$$

Define the training error

$$\text{Train} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \overline{Y})^2.$$

Let Y_{new} be an independent observation from the same population, and define

$$\text{New} = (Y_{\text{new}} - \overline{Y})^2.$$

(a) Show that

$$\mathbb{E}[\text{Train}] = \frac{n-1}{n}\sigma^2.$$

(b) Show that

$$\mathbb{E}[\text{New}] = \left(1 + \frac{1}{n}\right)\sigma^2.$$

(c) Explain why the expected training error is smaller, even though there is no question of choosing among complicated models.

(d) Verify the two formulas by simulation for $n = 10$, $\mu = 0$, and $\sigma = 1$. Use 20,000 repetitions and initialize the random-number generator once with seed 3373.

2 Empirical Study: KNN for the Iris Data

Open the file `data/iris.csv`. The first four columns are measurements and the final column gives the variety of iris.

Randomly split the observations into a development set and a final test set, each of roughly the same size. Use `train_test_split` with `test_size=0.5`, `random_state=3373`, and class stratification.

(a) Using only the development data, use five-fold stratified cross-validation to compare standardized KNN classifiers for $k \in \{1, 2, \ldots, 15\}$. Shuffle the folds and use `random_state=3373`. Plot the mean cross-validation misclassification error against k.

(b) Select k using the cross-validation results. Refit this classifier using all of the development data, then report its final test error and confusion matrix.

(c) Fit standardized logistic regression using the same development data. Report its final test error and confusion matrix.

(d) Comment briefly on the two classifiers. If you obtained a new independent test set from the same population, would you expect its error to be systematically larger, systematically smaller, or about the same as the observed final test error?

(e) Suppose instead that you had selected k by finding the smallest error on the final test set. Explain why reporting that same minimum as final performance would tend to be optimistic.

3 A Nonlinear Boundary Made Simple by One New Feature

The files `data/circular_train.csv` and `data/circular_test.csv` contain two predictors, X_1, X_2, and a binary response Y. The test file is a genuinely untouched final test set.

(a) Plot the training observations and describe the apparent decision boundary.

(b) Fit ordinary logistic regression using (X_1, X_2). Display its fitted decision boundary. If the boundary does not cross the plotted range, report its equation and say so.

(c) Define the new feature

$$R^2 = X_1^2 + X_2^2.$$

Fit logistic regression using R^2 as its predictor, and display the fitted boundary in the original (X_1, X_2) coordinates.

(d) Fit a standardized KNN classifier. Choose $k \in \{1, 3, 5, \ldots, 41\}$ using five-fold stratified cross-validation on the training data, with mean misclassification error as the criterion. Shuffle the folds and use `random_state=3373`.

(e) Compare final test error for the three fitted rules. Also compare final test log loss for the two logistic-regression rules.

(f) Explain why the improvement after introducing R^2 is usefully described as a change of representation. State one conclusion supported by this experiment and one broader conclusion that is not supported.

4 Lying with Linear Regression

It is often possible to get very misleading results by deliberately choosing a bad model with a certain structure. In this question, we will practice doing this in a simple setting. In case it was not clear: you should not actually do this in a real analysis. Seeing very misleading results can, however, help you learn to diagnose them.

Throughout the question, use at least 20 distinct X-values. Every chosen X-value is paired with the observed value

$$Y = \sin(X).$$

There is no measurement error. We will nevertheless fit the data to the usual linear regression model with unknown β_0, β_1, and σ.

(a) Find a collection of predictors $X_1, \ldots, X_n$ for which the reported 99-percent confidence interval for the slope β_1 is contained in $(-\infty, -0.95]$. Display the points and report the estimated slope and confidence interval.

(b) Find a collection of predictors $X_1, \ldots, X_n$ for which the reported 99-percent confidence interval for the slope β_1 is contained in $[0.95, \infty)$. Display the points and report the estimated slope and confidence interval.

(c) Would part (b) be possible if $[0.95, \infty)$ were replaced by $[100, \infty)$? Explain why or why not.

A complete proof for part (c) is not required, but will be considered for bonus marks. One useful fact is that sin is 1-Lipschitz.

5 Simulation Study: Post-Selection Inference

Generate $m = 80$ predictors and one response for $n = 100$ observations according to

```
rng = np.random.default_rng(3373)
X = rng.normal(size=(100, 80))
y = rng.normal(size=100)
```

There is no relationship between any predictor and the response.

(a) For each predictor, compute its sample correlation with y. Let j_1, j_2 be the indices with the two largest correlations in absolute value. Fit

$$Y = \beta_0 + \beta_1 X_{j_1} + \beta_2 X_{j_2} + \epsilon$$

using the same 100 observations. Report the two slope estimates, 95-percent confidence intervals, and p-values. Comment on the result.

(b) Use the first 50 observations only to select j_1, j_2, and use the remaining 50 observations only to fit the displayed model and carry out the two tests. Again report 95-percent confidence intervals and p-values, and comment on the difference.

(c) Repeat the complete simulation 200 times. Initialize the random-number generator once before the repetition loop; do not reset the seed inside the loop. For each of the two procedures, estimate the proportion of repetitions in which at least one of the two reported slope p-values is below 0.05.

(d) Explain why the first procedure produces apparently significant relationships so often, and why the second procedure restores the usual null interpretation of each individual p-value. Explain why the probability that at least one of two p-values is below 0.05 is not itself generally 0.05.

6 An Exact Bias–Variance Calculation for a Local Regression Rule

For $i = 1, \dots, n$, let

$$X_i = \frac{i}{n}, \qquad Y_i = X_i + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2).$$

The regression function is $f(x) = x$. To estimate $f(0) = 0$, use the unweighted k-nearest-neighbour rule

$$\widehat{f}_k(0) = \frac{1}{k} \sum_{i=1}^{k} Y_i.$$

(a) Compute the bias and variance of $\widehat{f}_k(0)$.

(b) Show that

$$\mathrm{MSE}(k) = \frac{(k+1)^2}{4n^2} + \frac{\sigma^2}{k}.$$

Identify the squared-bias and variance terms. State which becomes larger and which becomes smaller as k increases.

(c) Treat k as continuous and replace $k + 1$ by k. Find the resulting approximate minimizer and show that the optimal scale is $k \asymp n^{2/3}$ when σ is fixed.

(d) For $n = 200$ and $\sigma = 0.2$, compare $k \in \{1, 5, 15, 50\}$ using the exact formula. Verify the same pattern by simulation.

(e) Explain which features of this example make the calculation possible, and why it is not a universal theorem that every more flexible method must have larger variance.

7 Predicting a New Visit or Predicting a New Person?

The file `data/repeated_visits.csv` contains five visits from each of 500 people. The columns are:

- `person_id` and `visit_id`;
- one noisy feature called `signal`;
- six nearly constant features called `fingerprint_1`, ..., `fingerprint_6`;
- a binary response called `outcome`, which is the same for all visits from one person.

Do not use `person_id` or `visit_id` as predictors.

(a) Fit standardized logistic regression and standardized 1-NN using a random row-wise 50–50 training/test split. Use `train_test_split` with `test_size=0.5`, `random_state=3373`, and class stratification. Report test accuracy.

(b) Repeat the comparison using `GroupShuffleSplit`, with `person_id` as the group label. Use one split, `test_size=0.5`, and `random_state=3373`. Thus all five visits from a person must be on the same side of the split.

(c) Explain why the row-wise evaluation is especially favourable to 1-NN. Under the idealization that each of a test visit's four companion visits independently lands in training with probability 1/2, calculate the probability that at least one companion visit is in the training set.

(d) What prediction task does the row split approximate? What prediction task does the group split approximate?

(e) Suppose the system will be used for entirely new people. Which result should be used to compare the models? Suggest one additional check before deployment.

(f) Remove the six fingerprint features and repeat the two evaluations. Explain the change.